Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is a key difference between t-SNE and traditional multidimensional scaling? 1 / 1 point

- t-SNE is less computationally intensive
- t-SNE uses linear transformations
- t-SNE focuses more on local neighborhood structures
- t-SNE preserves global distances

✓ Nice work

t-SNE gives more weight to preserving the local neighborhood structure of the data, unlike traditional MDS which focuses on preserving global distances.

2. Why is t-SNE not typically used as a preprocessing step for clustering? 1 / 1 point

- t-SNE alters the original data distribution
- t-SNE requires excessive computational resources
- It does not preserve cluster density or separation
- It only works with numeric data

✓ Nice work

t-SNE does not preserve important global distance-based properties, including cluster density and cluster separation, which are often critical for effective clustering.

3. What is a key difference between UMAP and t-SNE in terms of dimensionality reduction capabilities? 1 / 1 point

- UMAP can preserve some global structure
- t-SNE can handle higher dimensional data more efficiently
- UMAP requires more computational resources
- t-SNE preserves more local structure than UMAP

✓ Nice work

Unlike t-SNE, UMAP is capable of preserving some aspects of the global structure of the data while also maintaining local neighborhood relationships, making it versatile for more applications, including use in machine learning pipelines.

4. What is an intuitive interpretation of the perplexity parameter in t-SNE? 1 / 1 point

- It's roughly an estimate of how many neighbors are near each data point
- It controls the algorithm's convergence rate
- It's roughly an estimate of the difficulty of separating clusters of similar points.

✓ Nice work

Correct. It can also be interpreted as a way to balance between local and global aspects of the data. The results of t-SNE can be very sensitive to the perplexity parameter so it important to explore multiple values.